

Beyond Permutation Symmetry: Symmetry, Redundancy, and Neural-Network Connectivity

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Abstract

The single-basin-modulo-permutation conjecture of Entezari et al. (2021) proposes that independently trained neural networks can often be connected by a low-loss linear path after permuting their hidden units. We test the limits of this explanation. Exhaustive enumeration of all hidden-unit permutations in minimal-width XOR networks yields a counterexample among gradient-trained solutions: pairs of successful endpoints can remain linearly incompatible under every permutation. We then consider a larger exact symmetry class. Building on the differentiable Sinkhorn re-basing method of Peña et al. (2023), we augment permutation alignment with function-preserving positive ReLU scaling. We also introduce a method-agnostic refinement that freezes the permutation returned by any alignment method and subsequently optimizes only the scales. On XOR, scale-aware alignment achieves barriers below the exhaustive permutation-only optimum, showing that its gains cannot be explained by finding better permutations. It also lowers barriers relative to permutation-only alignment across all evaluated VGG/CIFAR-10 and MLP/Fashion-MNIST architectures. Finally, we study training-stage connectivity by extending the same-stage analysis of Ainsworth et al. (2023) to checkpoints from two independent runs at unequal epochs, $n \neq N$. The worse-endpoint barrier, defined as the maximum interpolation loss in excess of the larger endpoint loss, is negligible early in training and becomes non-negligible only after a distinct training stage. Together, these results show that per-

mutation alone does not exhaust the function-preserving transformations relevant to interpolation and identify when excess interpolation barriers emerge during training.

1 INTRODUCTION

Are independently trained neural networks genuinely in different basins, or do they only appear separated because their hidden features use different coordinates? The question matters both for the geometry of neural-network loss landscapes and for operations such as weight averaging. A high-loss straight line between two parameter vectors is not conclusive: neural-network parameterizations are non-identifiable, and functionally equivalent representatives can have very different interpolation behavior.

The single-basin-modulo-permutation conjecture proposes that most stochastic gradient descent (SGD) solutions can be permuted into linear mode connectivity (LMC) (Entezari et al., 2021). Re-basing methods make this hypothesis operational and substantially reduce barriers in MLPs and convolutional networks (Ainsworth et al., 2023; Peña et al., 2023). Yet a residual barrier after alignment is ambiguous. An algorithm may have missed a suitable permutation, the permutation symmetry class may itself be too small, or the architecture may lack the representational freedom needed for a low-loss path. Large-network studies cannot cleanly distinguish these explanations because their permutation groups are combinatorial and alignment is approximate.

We isolate these mechanisms using XOR networks small enough for exhaustive permutation search, then test the resulting account in VGG networks. The minimal-width experiment first removes the search confound: some trained pairs remain classification-incompatible under every permutation. We reproduce this behavior with full-batch gradient descent (GD) and minibatch SGD. We then enlarge the exact symmetry class using positive ReLU scaling. Starting from the exhaustive

best permutation and optimizing scales alone identifies gains that cannot be attributed to better correspondence search. Finally, nonlinear path and width experiments separate representative choice from architectural redundancy: symmetry moves within an equivalence class, whereas extra units create additional routes through parameter space.

Our contributions are:

- We give an exhaustive counterexample to universal permutation-based LMC for trained networks. At minimal XOR width, 22/28 successful pairs fail classification-LMC after the best permutation under both full-batch GD and minibatch SGD.
- We introduce scale-aware alignment and show, after exhaustive permutation search, that positive scaling lowers the loss barrier for all 67 tested XOR pairs at widths 3, 5, and 7.
- We show an empirical transition in nonlinear connectivity between widths two and three, distinguishing representational redundancy from symmetry alignment. Scaling both endpoints does not overcome the width-2 failure, but improves nonlinear path optimization at larger widths for both polygonal and Bézier parameterizations.
- We show that scaling improves over permutation-only alignment on fixed VGG11/13/16/19 pairs trained on CIFAR-10.

2 PROBLEM FORMULATION AND RELATED WORK

Connectivity and interpolation barriers. For endpoints θ_A and θ_B , linear interpolation is $\theta(\lambda) = (1 - \lambda)\theta_A + \lambda\theta_B$, with $\lambda \in [0, 1]$. Following Entezari et al. (2021), the barrier is

$$B(\theta_A, \theta_B) = \sup_{\lambda \in [0, 1]} \left[C((1 - \lambda)\theta_A + \lambda\theta_B) - [(1 - \lambda)C(\theta_A) + \lambda C(\theta_B)] \right]. \quad (1)$$

Here $C : \Theta \rightarrow \mathbb{R}$ is a scalar cost assigned to a model. For gradient-based alignment, C must be differentiable and denotes binary or multiclass cross-entropy in our experiments. We use C to follow the re-basing literature and to distinguish the generic cost from a particular choice of loss. Classification error is used only as a reporting cost and is never optimized. The supremum is approximated on a finite grid. Nonlinear mode connectivity instead asks whether a general curve $\gamma : [0, 1] \rightarrow \Theta$ joins the endpoints at low loss. A curved path may avoid a region crossed by the straight segment, so nonlinear connectivity does not imply LMC (Draxler et al., 2018; Garipov et al., 2018).

Permutation re-basing. Relabeling a hidden layer and applying the inverse relabeling to the next layer preserves the network function; the same action reorders channels in convolutional networks. Weight matching finds layerwise permutations that reduce Euclidean distance, activation matching aligns representations on data, and differentiable methods optimize soft assignments (Ainsworth et al., 2023; Peña et al., 2023). These successes support a symmetry-aware view of loss landscapes, but a residual barrier obtained by a heuristic cannot establish that the symmetry class itself is insufficient.

Why linear connectivity matters. A curved connection establishes geometric connectedness, but a straight path is the relevant object for parameter averaging. Checkpoint averaging can improve generalization within one trajectory (Izmailov et al., 2019), and averaging related fine-tuned models can be effective without ensemble-time cost (Wortsman et al., 2022). Our focus is the geometric prerequisite of a low-barrier segment, rather than a claim that an aligned midpoint must outperform its endpoints.

3 SCALE-AWARE ALIGNMENT

Permutation is only one class of exact symmetry in neural networks. Another is the scaling symmetry induced by positive homogeneity: for ReLU, $\sigma(cz) = c\sigma(z)$ for $c > 0$, so the incoming weights and bias of a unit can be multiplied by c , provided that its outgoing weights are divided by c . Applying permutation and positive scaling together is therefore an exact symmetry, selecting a different parameter-space representative of the same endpoint function. REPAIR also uses channel-wise scaling and shifting, but applies it to interpolated activations and thereby changes the represented function (Jordan et al., 2023). Consequently, the intermediate networks obtained by linearly interpolating to that endpoint can have different losses, and hence a different barrier. This motivates studying connectivity under a richer symmetry class before attributing a residual barrier to genuinely distinct basins.

We extend Sinkhorn because it already expresses permutation search as a continuous, differentiable optimization, so continuous scale variables can be added and trained jointly under the same interpolation objective. This choice does not imply that other re-basing methods cannot incorporate scaling. Doing so in layerwise matching or straight-through methods would instead require modifying their matching objectives or coupling discrete permutation updates with continuous scale optimization, making the extension less direct.

Original Sinkhorn re-basing. For a network with L affine layers, let $h_\ell = \sigma(W_\ell h_{\ell-1} + b_\ell)$ for hidden layers and let P_ℓ relabel hidden layer ℓ . The induced action on adjacent weights and biases is

$$\widetilde{W}_\ell = P_\ell W_\ell P_{\ell-1}^\top, \quad \widetilde{b}_\ell = P_\ell b_\ell. \quad (2)$$

Because $P_\ell^{-1} = P_\ell^\top$ and ReLU commutes with permutation, Eq. 2 preserves the network function.

Sinkhorn re-basing replaces each discrete P_ℓ with a differentiable soft assignment (Peña et al., 2023). An unconstrained score matrix Z_ℓ is mapped to an approximately doubly stochastic matrix $S_\ell = \text{Sinkhorn}(Z_\ell/\tau)$, where the temperature τ controls its sharpness. It optimizes the layerwise scores using the relaxed action

$$\widetilde{W}_\ell^{\text{soft}} = S_\ell W_\ell S_{\ell-1}^\top, \quad \widetilde{b}_\ell^{\text{soft}} = S_\ell b_\ell. \quad (3)$$

Gradients through the Sinkhorn map are obtained by implicit differentiation. Since $S_\ell^\top$ is generally not S_ℓ^{-1} , the soft action need not preserve the endpoint function. Linear assignment projects each S_ℓ to a hard, function-preserving permutation for evaluation.

Scale-aware extension. For each hidden layer, we introduce log-scales u_ℓ and set $D_\ell = \text{diag}(e^{u_\ell})$. The exponential enforces the positivity required by ReLU homogeneity. Combining D_ℓ with a hard permutation gives the following function-preserving transformation:

$$\widetilde{W}_\ell = P_\ell D_\ell W_\ell D_{\ell-1}^{-1} P_{\ell-1}^\top, \quad \widetilde{b}_\ell = P_\ell D_\ell b_\ell. \quad (4)$$

Positive homogeneity propagates the scaled, permuted activation to the next layer, where the inverse factor cancels it; hence the transformed endpoint computes the same function.

During joint soft optimization, our implementation uses the corresponding relaxation

$$\widehat{W}_\ell = S_\ell D_\ell W_\ell D_{\ell-1}^{-1} S_{\ell-1}^\top, \quad \widehat{b}_\ell = S_\ell D_\ell b_\ell. \quad (5)$$

Only the diagonal scaling is inverted exactly; $S_{\ell-1}^\top$ remains the soft surrogate used by the original method. Projecting S_ℓ to P_ℓ recovers the exact action in Eq. 4.

We consider two variants. *Permutation + scale (joint Sinkhorn)* optimizes Z_ℓ and u_ℓ together. *Permutation + scale (fixed-permutation refinement)* first obtains and freezes a hard permutation, then optimizes u_ℓ alone. On XOR, the frozen permutation can be the exhaustive optimum, making the incremental contribution of scale identifiable. A quadratic penalty on u_ℓ controls the unbounded scaling symmetry and discourages numerically ill-conditioned representatives. Extending the standard re-basin notation $\pi_P(\theta_B)$, let $\pi_{Z,u}(\theta_B)$ denote the endpoint transformed by the current soft assignments and

scales. Following the original Sinkhorn method (Peña et al., 2023), we use the interpolation cost

$$C_\lambda(Z, u; \theta_A, \theta_B) = C((1 - \lambda)\theta_A + \lambda\pi_{Z,u}(\theta_B)). \quad (6)$$

Fixing $\lambda = 0.5$ gives the midpoint objective; drawing $\lambda \sim \mathcal{U}(0, 1)$ independently at each update gives the random-point objective. The midpoint is inexpensive and often the most vulnerable location, whereas random sampling covers the path across updates. We minimize Eq. 6 plus $\lambda_s \sum_\ell \|u_\ell\|_2^2$, where the sum is over hidden layers. Freezing a hard permutation removes Z from the optimization. We report barriers only after projecting the soft assignments to hard permutations. Experiment-specific objectives and grids are given in Appendix A.

4 EXPERIMENT I: PERMUTATION IS NOT ENOUGH

Question. When permutation-based LMC fails, did the alignment algorithm fail, or does no permutation work? We answer this in the smallest nontrivial setting: a one-hidden-layer ReLU network of width $h = 2$ trained on the four XOR examples. There are only $2! = 2$ hidden-unit permutations, so the symmetry class can be searched exactly.

Training and evaluation. Models are initialized independently and retained only if they reach 100% endpoint accuracy and low binary cross-entropy. Our primary protocol uses full-batch GD. As a robustness check against the interpretation that the effect is specific to deterministic optimization, we repeat the experiment with minibatch SGD using batch size two and reshuffling each epoch. For each protocol, eight successful endpoints give 28 pairs. We enumerate identity and swap and evaluate classification error along each interpolation.

Result. For both training protocols, only 6/28 pairs preserve 100% accuracy along the best-permuted segment; the remaining 22/28 incur interpolation error under both possible permutations. Figure 1 shows a successful pair and a representative full-batch failure. In the failure, both endpoints correctly implement XOR and have similar decision boundaries, but no hidden-unit relabeling makes their straight interpolation compatible. Because the search is exhaustive, this result cannot be attributed to weight matching, Sinkhorn relaxation, or alignment initialization.

The conclusion is deliberately scoped. These trained endpoints refute a universal claim that permutation-based LMC follows automatically from SGD or GD. They do not contradict the sufficiently-wide,

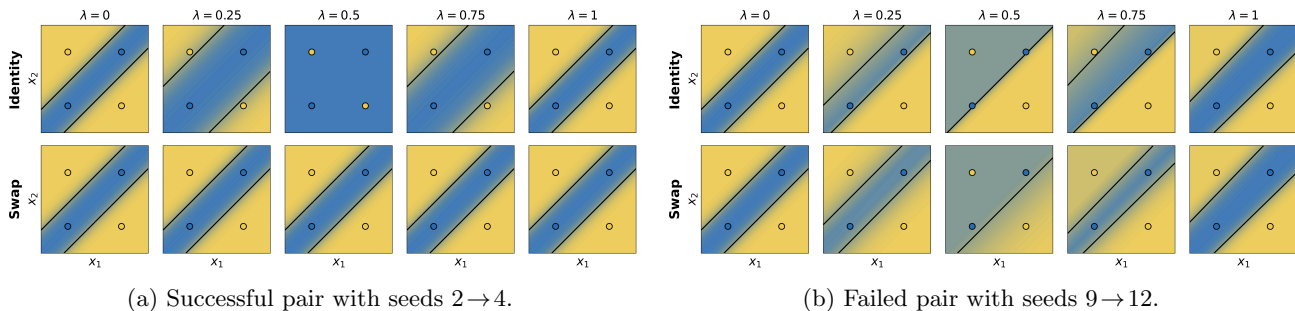


Figure 1: Decision boundaries along linear interpolation paths for two width-2 XOR endpoint pairs. Columns show interpolation values λ ; rows compare the identity and swap permutations.

high-probability form of the single-basin-modulo-permutation conjecture. They also qualify the interpretation of the hand-constructed counterexample of Ainsworth et al. (2023). Ainsworth et al. used adversarial, non-SGD endpoints to show that architecture alone does not guarantee permutation-based LMC, and argued that the empirical prevalence of LMC may therefore reflect an implicit bias of SGD. Our exhaustive result shows that this bias is not absolute: standard minibatch SGD and full-batch GD can themselves produce successful solutions for which no permutation yields LMC.

5 EXPERIMENT II: A RICHER SYMMETRY EXPLAINS MORE LMC

Question and setup. Does positive scaling improve compatibility once permutation-search error has been removed? We train full-batch GD XOR networks at widths $h = 3, 5, 7$, giving 21, 36, and 10 valid endpoint pairs. For each pair we enumerate all $h!$ permutations and select the one with the smallest sampled loss barrier. We then freeze the selected permutation and optimize only positive scales. Final barriers use 61 interpolation points. We also report raw interpolation and practical Sinkhorn baselines. Additional optimization details appear in Appendix A.

5.1 Width and permutation alignment

Increasing width changes the result. In the initial exhaustive sweep over $h \in \{2, 3, 4, 5\}$, the number of pairs maintaining 100% accuracy along the best-permuted interpolation rises steadily; at $h = 5$, every evaluated pair maintains perfect accuracy. Perfect accuracy does not mean zero loss barrier: midpoint confidence can still fall relative to the endpoints.

This width trend is consistent with Git Re-Basin, which empirically found lower post-alignment barriers for

wider VGG and ResNet models (Ainsworth et al., 2023). Our purpose here is not to re-establish that trend, but to measure it under exhaustive permutation search and then isolate the additional effect of scaling. Table 1 reports all aggregate results. The exhaustive-permutation mean falls from 0.357 at $h = 3$ to 0.023 at $h = 7$; practical Sinkhorn remains above the exhaustive result at each width, so failure of the heuristic alone cannot establish that no suitable permutation exists.

Variability across pairs. The standard deviations in Table 1 describe variation across endpoint pairs, not uncertainty in the mean. Barriers are nonnegative and strongly right-skewed: some independently trained pairs are almost connected, whereas others remain much harder. For example, at $h = 5$ the joint Sinkhorn+scale barrier has median 0.011 but ranges up to 0.692, yielding 0.074 ± 0.149 . The summaries are descriptive because pairs that share an endpoint are not statistically independent.

Scale beyond exhaustive permutation. The decisive comparison is between exhaustive permutation and the same frozen permutation followed by scale refinement. Refinement lowers the mean barrier by 31–78% across widths and lowers it on all 67 evaluated pairs. Since all permutations have already been enumerated and the selected permutation is held fixed during refinement, the comparison isolates the additional rescaling freedom. For these endpoints, rescaling reaches functionally equivalent parameterizations with lower interpolation barriers than are attainable by permutation alone.

Joint optimization is more mixed at the narrowest width. It achieves a lower mean barrier than exhaustive permutation at $h = 5$ and $h = 7$, but not at $h = 3$. Its gap to fixed-permutation scale refinement is therefore an optimization gap: the richer symmetry class contains better representatives even when a from-scratch solver does not reliably find them.

Alignment method	$h = 3$	$h = 5$	$h = 7$
No alignment	0.780 ± 0.610	0.608 ± 0.465	0.331 ± 0.184
Permutation only (Sinkhorn)	0.444 ± 0.325	0.189 ± 0.162	0.063 ± 0.044
Permutation only (exhaustive)	0.357 ± 0.309	0.111 ± 0.118	0.023 ± 0.015
Permutation + scale (joint Sinkhorn)	0.428 ± 0.380	0.074 ± 0.149	0.009 ± 0.009
Exhaustive permutation + scale refinement	0.247 ± 0.283	0.025 ± 0.044	0.005 ± 0.003

Table 1: XOR loss barrier across endpoint pairs (mean $\pm$ standard deviation). “Exhaustive permutation + scale refinement” selects the exhaustive-best permutation and then optimizes only positive scales while holding that permutation fixed. Lower is better.

Appendix Figure A.2 shows the corresponding mean interpolation profiles. The curves visualize the width trend and method ranking summarized more completely by Table 1.

6 EXPERIMENT III: WIDTH AND ENDPOINT SCALING IN NONLINEAR PATH FINDING

Question and setup. Residual linear barriers need not indicate disconnected low-loss sets. We therefore optimize the polygonal and quadratic Bézier paths used by Garipov et al. (2018). For endpoints θ_A, θ_B and one trainable interior control point v , their one-control-point parameterizations are

$$\phi_v^{\text{poly}}(\lambda) = \begin{cases} 2(\lambda v + (\frac{1}{2} - \lambda)\theta_A), & 0 \leq \lambda \leq \frac{1}{2}, \\ 2((\lambda - \frac{1}{2})\theta_B + (1 - \lambda)v), & \frac{1}{2} \leq \lambda \leq 1, \end{cases} \quad (7)$$

and

$$\phi_v^{\text{Bézier}}(\lambda) = (1 - \lambda)^2\theta_A + 2\lambda(1 - \lambda)v + \lambda^2\theta_B. \quad (8)$$

The polygonal path consists of two line segments meeting at v ; the Bézier path bends smoothly under its influence. We keep the endpoints fixed and optimize the control points to minimize the maximum loss over 31 sampled path locations, using up to ten interior points. To test whether width affects available routes rather than only alignment, we apply these paths to independently trained XOR endpoints. At widths $h \in \{2, 3, 5, 7\}$, we also compare a one-control-point path with joint optimization of that path and independent positive scales at both endpoints. Scaling preserves the endpoint functions; no permutation alignment is applied in this comparison.

Result. At $h = 2$, extensive searches find low-loss nonlinear paths only between essentially identical endpoints; additional bends do not recover nontrivial connections. At $h = 3$, all studied endpoint pairs are connected by low-loss paths. Most are found directly; for one difficult pair, concatenating low-loss paths through

a third trained mode establishes connectivity. Figure A.1 in the appendix contrasts representative outcomes.

This result is empirical, not a proof that no width-2 path exists outside the tested families. The only two successful width-2 pairs, seeds 2–9 and 5–12, are nearly duplicate parameter solutions: their Euclidean endpoint separations are only about 0.6% of the corresponding parameter norms, and both path families attain barriers at numerical zero. They therefore do not provide evidence of a nontrivial connection between distinct width-2 solutions.

A plausible explanation for the remaining failures is that width 2 is capacity-tight for XOR. The nontrivial endpoint pairs have different decision-boundary geometries, each constructed from only two ReLU boundary pieces. Within the tested path families, continuously transforming one geometry into the other with no spare hidden unit requires an intermediate arrangement that misclassifies at least one XOR point. After adding a third unit, all studied pairs admit low-loss paths, consistent with the extra unit providing the temporary representational freedom needed during this transformation. This supports a distinction that endpoint alignment alone cannot make: symmetry selects among equivalent representatives at fixed capacity, whereas width changes the representational resources available along a path.

Endpoint-scaling result. Endpoint scaling does not remove the minimal-width obstruction. With one interior control point, both polygonal and Bézier searches find low-loss connections for only 2/28 width-2 pairs, with or without jointly optimized endpoint scales. Thus, rescaling helps the optimizer choose representatives but does not manufacture a route that the narrow architecture cannot express within the tested path families.

At larger widths, scaling nevertheless improves path finding. For polygonal paths, it lowers the mean barrier by 1.3%, 4.7%, and 94.9% at widths 3, 5, and 7, respectively. Quadratic Bézier paths show the same

qualitative higher-width pattern: little change at width 3, a modest reduction at width 5, and a reduction of more than 90% at width 7. The two path families also give identical low-loss connection counts at every width. At width 2, their mean-barrier details differ slightly—scaling improves the polygonal mean but worsens the Bézier mean—yet neither finds additional low-loss connections. Scaling is therefore useful for both linear alignment and nonlinear path optimization, while remaining insufficient to overcome the width-2 capacity limitation. Figure 2 summarizes the polygonal-path comparison.

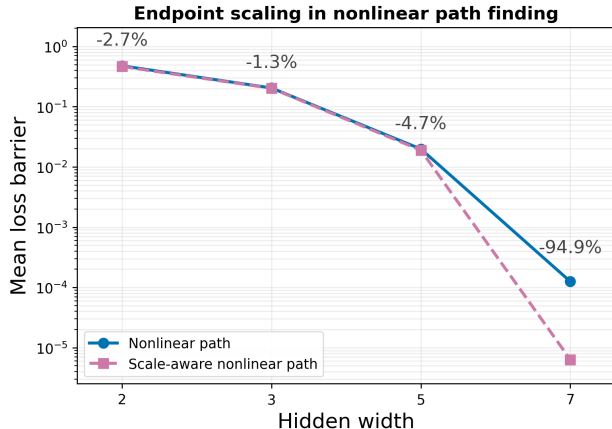


Figure 2: Mean nonlinear-path loss barrier by hidden width for polygonal paths with one interior control point, optimized from raw endpoints with or without independent positive scaling of both endpoints. Percentage annotations report the relative change under endpoint scaling.

7 EXPERIMENT IV: VGG/CIFAR-10 VALIDATION

Setup. We use fixed, independently trained seed-0/seed-1 pairs of VGG11, VGG13, VGG16, and VGG19. Alignment uses a deterministic 1000-example calibration batch. Besides raw interpolation, we test Sinkhorn permutation alignment, a joint variant that optimizes the soft permutation and positive scales together, and a sequential variant that freezes the hard Sinkhorn permutation before refining only its scales. Test-set barriers are evaluated after hard projection.

Result. Figure 3 reports test-loss barriers for the four architectures. Sinkhorn permutation alignment reduces the raw barrier for every model, and allowing positive scales improves it further in every case. The largest gain is on VGG11, where the best scale-aware barrier is about 3.5 times lower than with permutation alignment alone. Neither optimization order dominates: joint op-

timization is better for VGG11 and VGG19, whereas sequential scale refinement is better for VGG13 and VGG16. This echoes XOR: the symmetry class contains better representatives, but joint nonconvex optimization does not always realize the full gain. Substantial barriers also remain for deeper VGGS, so permutation and scale together are not a complete account of LMC.

The VGG results should be read as a transfer check of the controlled XOR finding. Exhaustive permutation search is impossible, only one endpoint pair is used per architecture, and hyperparameters are selected per method. Nevertheless, the consistent direction of change across four architectures supports the practical relevance of scaling beyond the toy setting.

8 LINEAR TRAINING-STAGE CONNECTIVITY

Standard mode-connectivity experiments compare converged solutions, but this does not reveal when incompatibility between independent training runs first appears. Moreover, an intermediate checkpoint need not be a local minimum, so we refer to this question as *training-stage connectivity* rather than mode connectivity. Let $\theta_A^{(n)}$ and $\theta_B^{(m)}$ denote checkpoints from two independently initialized VGG11 models after n and m completed epochs. We ask how their interpolation geometry changes as either endpoint advances through training, and whether any emerging barrier is explained by exact parameter symmetries.

We train six no-batch-normalization VGG11 models on CIFAR-10 and group them into three independent pairs. For each run we retain checkpoints at epochs $\{0, 1, 2, 3, 5, 10, 20, 30, 60, 100, 150, 200\}$, where epoch zero is the untrained initialization and is analyzed separately from trained checkpoints. The complete design evaluates the ordered 12×12 matrix $\theta_A^{(n)} \leftrightarrow \theta_B^{(m)}$. Its diagonal measures same-stage compatibility, while the final row and column compare an intermediate model to a converged model in both independent-run directions.

For every checkpoint pair, we compare raw interpolation, weight matching, Sinkhorn permutation alignment, and variants that augment either alignment with function-preserving positive ReLU scaling. The first seed pair is used to select, separately for each matrix cell, the best permutation-only method and the best method overall using validation loss. These choices and their hyperparameters are then frozen and evaluated on two held-out seed pairs. Alignment optimization uses the same 45,000 training examples as endpoint training; final profiles use a fixed stratified 10,000-example training subset and the full CIFAR-10 test set. Test data

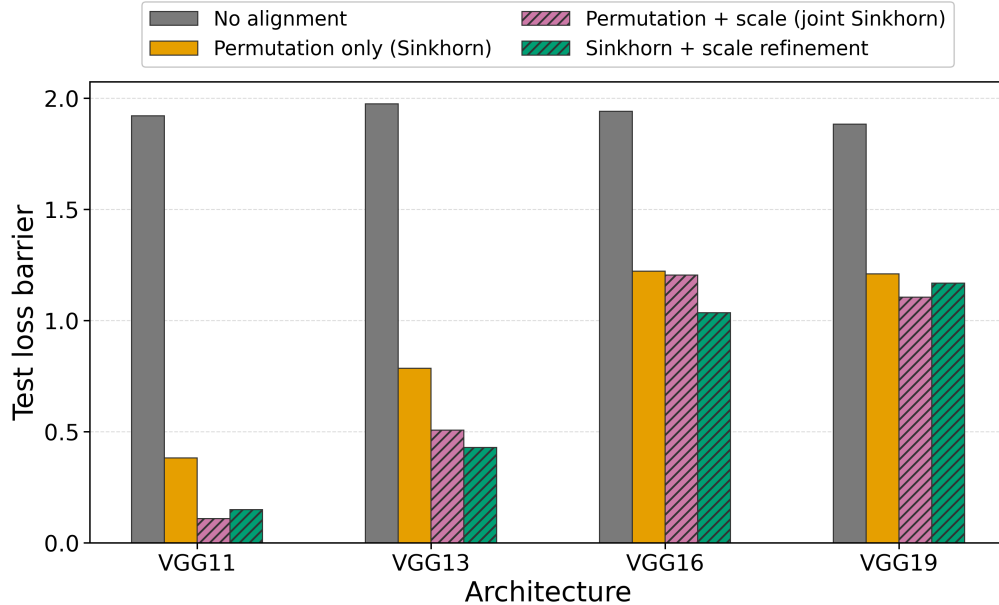


Figure 3: Maximum test-loss barrier for independently trained VGG models on CIFAR-10. Both scale-aware approaches improve on Sinkhorn permutation alignment for all four fixed endpoint pairs, although joint and sequential optimization trade places across architectures.

are not accessed during alignment or method selection.

We report the chord barrier from Eq. 1, denoted B_{chord} , together with

$$B_{\text{worse}} = \sup_{\lambda \in [0,1]} C(\theta(\lambda)) - \max\{C(\theta_A), C(\theta_B)\}. \quad (9)$$

The former measures deviation from the line joining the two endpoint costs and maintains comparability with re-basing work. The latter asks whether the path ever performs worse than its worse endpoint, which is especially useful when $n \neq m$ and endpoint losses differ substantially. Both quantities are computed for cross-entropy and classification error. This design separates three questions: when linear incompatibility emerges, how much permutation removes, and how much additional mismatch is explained by positive scaling.

9 DISCUSSION AND LIMITATIONS

Symmetry and redundancy play different roles.

The experiments support a two-part account of apparent basin separation. Permutation removes discrete coordinate mismatch, and positive scaling moves continuously along the same exact symmetry orbit. Neither changes the endpoint function. Width, by contrast, changes the architecture’s representational freedom and can create low-loss routes unavailable to a narrower network. Curved paths further relax the requirement that this route be the straight segment. These mechanisms interact, but they are not interchangeable: a

better representative cannot manufacture absent capacity, and additional capacity does not identify the right representative by itself. The nonlinear experiments make this interaction explicit. Scaling both endpoints lowers path-optimization barriers once sufficient width is available, but it does not produce additional low-loss connections at width 2.

The minimal-width experiment also sharpens the interpretation of Ainsworth et al. (2023). Their hand-constructed counterexample shows that architecture alone does not guarantee permutation-based LMC, and they suggest that LMC may be an artifact of SGD. Our result rules out a broad reading under which standard gradient-based training itself guarantees alignable endpoints: neither minibatch SGD nor full-batch GD prevents permutation-incompatible successful solutions. It does not rule out an SGD bias toward compatibility in wider or typical modern networks. Indeed, the rapid fall in the exhaustive permutation barrier with width is consistent with the sufficiently-wide, high-probability form of the single-basin-modulo-permutation conjecture (Entezari et al., 2021; Ainsworth et al., 2023).

Optimization versus existence. The exhaustive protocol is central to this interpretation. Without it, a high barrier after Sinkhorn or weight matching could always be ascribed to search failure. Conversely, the gap between joint Sinkhorn and fixed-permutation scale refinement warns against treating a failed continuous relaxation as evidence that a good functionally equiva-

lent representative does not exist. Future work should distinguish the expressivity of a transformation class from the solver used to search it.

Implications for alignment studies. A residual barrier should be interpreted in stages. Search error must first be separated from limitations of the transformation class. If permutation is insufficient, richer exact symmetries should be tested before inferring distinct basins. Linear and nonlinear paths should then be distinguished, and width should be treated as a change in representational resources rather than merely a larger matching problem.

Limitations. The exact conclusions come from small one-hidden-layer XOR networks. The minibatch and full-batch protocols are robustness replications, not a controlled estimate of stochasticity: with the same epoch count, batch size two performs more parameter updates than full-batch GD. The VGG study is non-exhaustive, uses a single pair per architecture, and relies on method-specific hyperparameter selection. Our nonlinear path searches can certify found paths but cannot certify nonexistence. The scale-aware soft transformation is exactly function preserving only after projection; during Sinkhorn optimization it is a relaxation. Finally, we consider ReLU networks, for which positive homogeneity holds, and do not address normalization layers, residual architectures, attention symmetries, or approximate functional equivalences. Larger multi-seed studies and comparisons with activation repair are needed before drawing broad architectural conclusions.

10 CONCLUSION

Permutation symmetry is an important but incomplete explanation of neural-network connectivity. Exhaustive XOR search shows that trained endpoints can remain incompatible under every permutation, for both minibatch SGD and full-batch GD. Positive scaling improves even the exhaustive best permutation, and it also lowers optimized nonlinear-path barriers at larger widths. However, scaling does not recover low-loss polygonal or Bézier connections for the difficult width-2 pairs. The width-dependent nonlinear experiment therefore shows that exact symmetries do not subsume representational constraints. The same linear-alignment benefit of scaling transfers to fixed VGG/CIFAR-10 pairs. The central distinction is therefore simple: symmetry determines which representatives we compare; redundancy determines which paths are available.

AI Use Statement

Generative AI tools were used to brainstorm presentation choices, improve the organization and readability of draft text, and assist with programming tasks including debugging, refactoring, plotting, and $\text{\LaTeX}$ editing. They were not used to generate experimental data, select the reported results, or make the final scientific interpretations. The research questions, experimental design, implementation decisions, analysis, and conclusions were determined by the authors. All AI-assisted text and code were reviewed and checked before inclusion. The authors take responsibility for the final content, claims, citations, and artifacts. This statement must be reconciled with the complete author team’s actual use before submission.

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CHECKLIST

For each question, the selected answer is shown in brackets, followed where useful by a short justification.

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes; Sections 2–3 and Appendix A.]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [No; empirical behavior rather than asymptotic complexity is the focus.]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [No.]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Not Applicable; no formal theorem is claimed.]
 - (b) Complete proofs of all theoretical results. [Not Applicable.]
 - (c) Clear explanations of any assumptions. [Yes; the positive-homogeneity assumption is stated in Section 3.]
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No; experimental settings are included, but an anonymized code release is not yet linked.]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes; Appendix A.]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes; Eq. 1, Table 1, and the limitations in Section 7.]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [No.]
 - (a) Citations of the creator If your work uses existing assets. [Yes.]
 - (b) The license information of the assets, if applicable. [No.]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable.]
 - (d) Information about consent from data providers/curators. [Not Applicable; only synthetic XOR and the public CIFAR-10 benchmark are used.]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable.]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable.]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable.]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable.]

Beyond Permutation Symmetry: Symmetry, Redundancy, and Neural-Network Connectivity: Supplementary Materials

A REPRODUCIBILITY DETAILS

A.1 XOR training and exhaustive alignment

The XOR set is $\{(0,0), (0,1), (1,0), (1,1)\}$ with labels $\{0,1,1,0\}$. Models have one ReLU hidden layer, biases, and one output logit. The primary endpoint protocol uses full-batch GD (implemented with the SGD optimizer) at learning rate 0.05 for at most 5000 epochs. **ReduceLROnPlateau** uses factor 0.5, patience 200, threshold 10^{-4} , and minimum learning rate 5×10^{-4} . Training stops when loss has not improved by 10^{-4} for 500 epochs. For each retained pair, all $h!$ permutations are evaluated. Interpolation selection prioritizes classification error in the width-2 diagnostic and loss barrier in the scale-aware study.

The width-2 robustness replication uses true minibatch SGD with batch size two and reshuffles the four XOR examples every epoch. It otherwise uses the same initialization range, learning-rate schedule, stopping rule, maximum of 5000 epochs, accepted seeds, and exhaustive identity/swap evaluation. Eight successful endpoints produce 28 pairs in each protocol. Matching epoch counts does not match optimizer-update counts: the minibatch protocol makes two updates per epoch and is therefore a robustness comparison, not an isolated causal test of stochastic gradient noise. The width-3/5/7 scale-aware study uses the primary full-batch-GD endpoints.

For nonlinear paths, both quadratic Bézier curves and polygonal chains are optimized for 1500 steps at learning rate 0.05, with no weight decay and 31 path samples. The default path has one trainable interior point; harder pairs use up to ten. The endpoint-scaling comparison uses one interior point, raw endpoints without permutation, and independently optimized positive scales at both endpoints. Path parameters and endpoint log-scales are optimized jointly with Adam for 1500 steps at learning rate 0.05; the scale penalty is 10^{-4} , and final barriers are evaluated on 501 path locations.

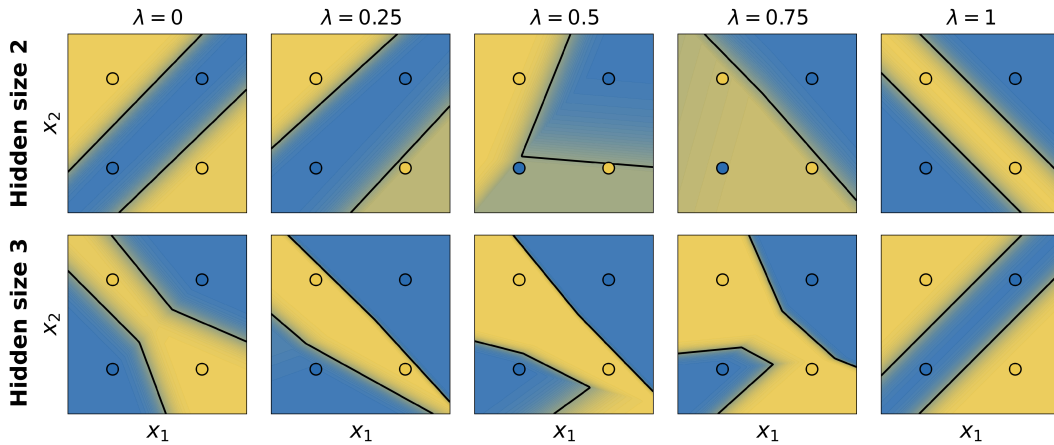


Figure A.1: Nonlinear XOR paths. *Hidden size 2*: a representative pair for which optimizing a Bézier path does not remove the high-loss region. *Hidden size 3*: a low-loss path connects endpoints with different decision boundaries. Rows visualize predictions along the learned path as a function of λ .

A.2 XOR scale-aware optimization

At each optimization step, XOR alignment evaluates binary cross-entropy at 31 evenly spaced interpolation points. The implementation subtracts the mean endpoint loss from each value and backpropagates the largest resulting excess; thus, the path-sample contribution to the gradient comes from the currently worst grid point rather than an average over all 31 locations. Final reported curves are evaluated on a separate 61-point grid. The default permutation only (Sinkhorn) run uses Adam for 500 steps, learning rate 0.01, temperature 1, 20 Sinkhorn iterations, and identity strength 1. Scale refinement uses Adam for 500 steps, learning rate 0.01, and log-scale regularization 10^{-3} . Joint optimization uses the same default budget. If a default run misses its target, a bounded search over learning rates, temperatures, and regularization is used; settings are selected per pair. The scale target is barrier $\epsilon = 0.01$, while permutation-only Sinkhorn targets that pair’s exhaustive barrier.

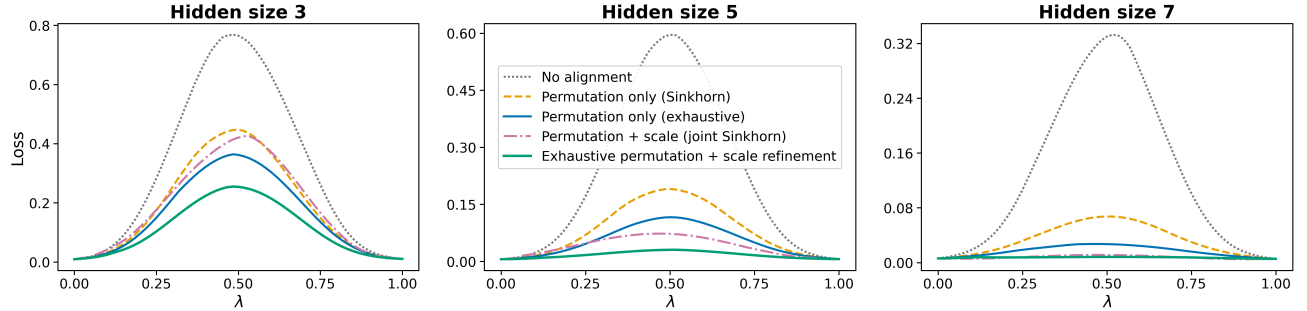


Figure A.2: Mean XOR interpolation-loss profiles at different hidden widths, comparing four alignment variants with raw interpolation. At each λ , loss is averaged across endpoint pairs.

A.3 VGG experiments

For the scale-aware comparison, fixed seed-0/seed-1 endpoint pairs are used for VGG11/13/16/19. Alignment uses a deterministic batch of 1000 images, 20 Sinkhorn iterations, and midpoint loss. Separate bounded sweeps select the number of optimization steps, learning rate, temperature, and scale regularization for each architecture and method. The permutation-then-scale baseline freezes the projected hard permutation and optimizes only log-scales.